

Batter's Swing Speed as Exogenous and Endogenous Variable. CSAS 2025 Data Challenge

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Abstract

The report analyzes the swing speed of the batter on fastball pitches both as an exogenous and endogenous variable. In the exogenous analysis, the batter randomly modifies the swing speed as an independent variable. In the endogenous analysis, the variable is treated as an indicator of the batter's strategy given the environment.

The main implication of the exogenous analysis is that the higher swing speed is associated with a higher probability of hitting the ball, suggesting that there is no optimal swing speed inside the range that maximizes such probability. As for the endogenous analysis, the main predictors of the swing speed are the score at bat (balls and strikes) and the direction of the pitch (i.e., where the ball crosses the strike zone). At the same time, the majority of the swing speed variance remains unexplained, which gives room for further analysis of the swing speed as both endogenous and exogenous variables.

1 Introduction

I cannot emphasize how wrong I was in thinking that baseball is a pretty simple game. After learning the game's rules this fall, I realized how complex and detailed the rules and strategies of the game are. This explains the abundance and variety of baseball data and statistical models available. At the same time, the data availability is still constantly growing. For example, the bat tracking data became publicly available only [this year](#) meaning that the analysis of such data is yet to be done.

This report analyzes the swing speed of batters on fastball pitches. The analysis is done in two parts. The swing speed is treated as an exogenous variable in the first part. I assume that the batter randomly chooses the swing speed. Given that, I analyze how the swing speed affects the probability of hitting the ball into play. Another goal in this part is to find out whether there is an optimal swing speed within the batter's range, which maximizes the probability of hitting the ball. This would imply that sometimes the batter is better off swinging the bat slower to improve the quality of the hit.

The swing speed is treated as an endogenous variable in the second part of the analysis. In this case, the variable can be interpreted as the indicator of the batter's cautiousness or confidence. The main goal is to find out the predictors, which affect the swing speed the most, and to what extent the environment can influence the batter's decision to swing the bat faster or slower.

The remainder of the report is structured as follows. In Section 2, I describe the dataset I use for the analysis. In Section 3, I describe the data preparation for the modeling. In Section 4, I present the exogenous analysis, and the endogenous one is presented in Section 5. In Section 6, I conclude. Finally, I state the directions for further work in Section 7.

2 Data

The data is provided by the [CSAS 2025](#) Data Challenge. It is pitch-level data for the MLB 2024 season and contains 701557 rows from the matches from 4/2/2024 to 10/30/2024. Each row represents the pitch and contains 113 variables that describe various characteristics of the pitch, including the game, date, inning, current score in runs, balls and strikes (and outs), players on the field (and bases) on each position, various characteristics of the pitch (speed, trajectory, pitch type, spin, etc), and characteristics of the outcome of the pitch (whether the batter swung, hit, how hard he hit, whether he hit the ball into play, the outcome of the pitch, etc).

The main variable in this analysis is `bat_speed` which is the swing speed of the batter. The speed is measured at the sweet spot of the bat (around 6 inches from the end of the bat) in mph. Note that I have not found the information about at which moment this speed is measured (i.e. at the moment of contact with the ball, or at the moment it crosses the strike zone), and whether it's some average speed or maximum speed. This information is yet to be found. Currently, I assume that the speed is at least measured at the same moment for each pitch.

3 Data Cleaning

First, this report considers the swing speed once the decision to swing was made. The analysis of the decision to swing (swing decision score models) is out of the scope of this analysis. Thus, I filtered out all the rows where the swing was not made. Bunts and 10% of the slowest swings for each batter are also filtered out. The reason for doing that is to leave only 'proper swings' in the data. By 'proper swings,' I mean the swings where the batter firmly decides to swing the bat. The bunt is another type of action during the pitch that is more suitable for the swing decision score models. The majority of the 10% of the slowest swings contain very slow swings, where the batter changes his mind but already makes enough movement to be counted as a swing. About 45% of pitches are swung, so after removing 10% of the slowest swings, 288630 rows are left in the data.

Another important decision is to consider only fastballs in this report. The swing speed and the outcome of at-bat heavily depends on the pitch. There are several types of pitches, each of which has unique characteristics and affects the batter's performance uniquely. Thus, I think that each pitch type should be analyzed separately. I decided to analyze the fastballs first. The trajectory of this pitch type is pretty straight, so these pitches are well approximated by the pitch speed, location of the ball, and the horizontal and vertical movement of the ball. At the same time, this pitch type is the most popular. 35% of pitches are fastballs. As for the other pitch types, the more detailed characteristics of the trajectory of the ball are needed, so one should be more careful when dealing with the other pitch types. The analysis of the other pitch types can be a part of the further work.

Finally, I will mainly look at the swing speed normalized by subtracting the batter's mean from the swing speed. To ensure the demeaning process is robust, I sacrifice 2% of the remaining data by filtering out the batters with less than 30 fastballs properly swung. After that, additional filtering out of the data that was not correctly measured resulted in 92903 rows being left for the modeling. The final cleaned data contains 2354 games and 509 players who have properly swung at least 30 fastballs in the MLB 2024 season. This means that the cleaned dataset contains, on average, 182.5 swings per batter. I use this cleaned dataset in both exogenous and endogenous analyses.

4 Swing Speed as an Exogenous Variable

In this section, I analyze how the swing speed affects the batter's performance. The probability of hit into play (i.e., hitting the ball into the fair zone) is used as the metric for such quality. I assume that the swing speed is exogenous, i.e. the batter randomly chooses the swing speed. This assumption is needed to make some causal statements and suggest the optimal swing speed. This assumption is challenged in the next section, where the swing speed is treated as an endogenous variable.

In Section 4.1, I present the models and the results. Before the modeling, I did some exploratory data analysis which helped me to create the variables used in the models. You can find the main results (visualizations and comments) from this exploratory data analysis in Appendix A.

4.1 Modelling and Results

4.1.1 Simple aggregate model. First, let me introduce the simple model, which was run on all the data simultaneously. The only reason to run this simple model (with potential bias) is its easy interpretability.

```
glm(formula = TARGET ~ bat_speed_norm * release_speed_norm +
     bat_speed_norm_sq, family = binomial, data = d1)
```

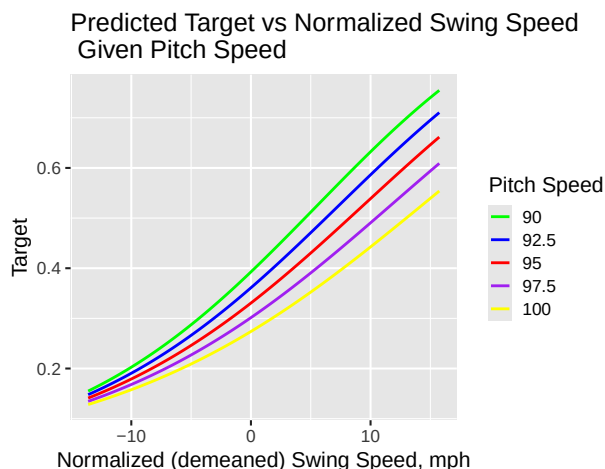
	Variable	Coefficient	SE	pvalue
1	(Intercept)	-0.68	0.0084	0
2	bat_speed_norm	0.085	0.002	0
3	release_speed_norm	-0.054	0.0029	2.3e-77
4	bat_speed_norm_sq	0.00022	0.00036	0.54
5	bat_speed_norm:release_speed_norm	-0.0023	0.00079	0.0031

The coefficients of this model are:

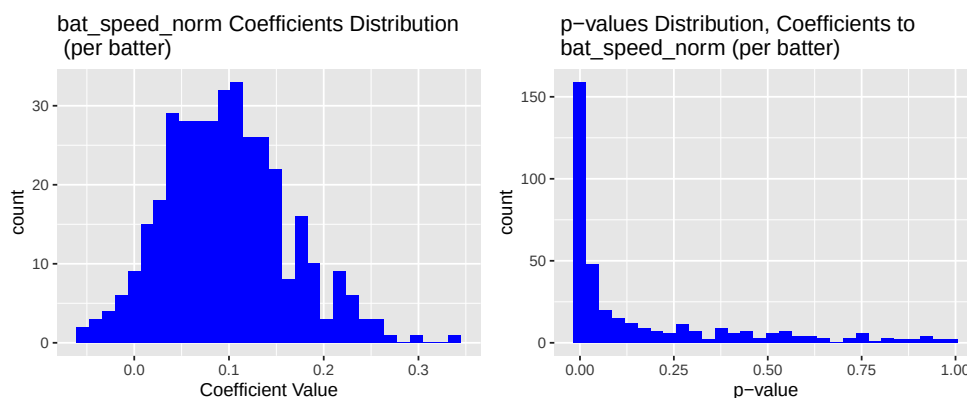
- bat_speed_norm: swing speed normalized by subtracting the batter's average swing speed (in mph)
- release_speed_norm: pitch speed normalized by subtracting aggregate average pitch speed in the dataset (in mph)
- bat_speed_norm_sq: square of the normalized swing speed
- bat_speed_norm:release_speed_norm: interaction term between normalized swing speed and normalized pitch speed

Interpretation of the coefficients: Note that this is the logit model for the binary outcome, not a linear regression. Thus, the coefficients are not directly interpretable. Still, the signs of the coefficients can be interpreted. If the ball is pitched at the average speed (IF WE FIX THE PLAYER), the batter's increase in swing speed is associated with the increase in the probability of hitting the ball into play. The effect gets slightly smaller if the pitch is thrown at a higher speed, but the effect never turns positive in the reasonable range of pitch speed (<105mph). The effect of the pitch speed is negative no matter what the swing speed is relative to the batter's average. The coefficient to the squared normalized swing speed is insignificant, so there is no evidence of the existence of the optimal swing speed.

Below, you can see the association between normalized swing speed and TARGET for different values of pitch speed.



4.1.2 Simple models for each player separately. I ran the model above on each player separately. I considered only 370 out of 509 players with more than 100 swings. There is not enough data for the other players to fit the model with 5 variables. It turns out that almost for any player, this simple model has too many variables to make any conclusions. The only significant result is a pretty stable positive coefficient to the normalized swing speed, suggesting that normalized swing speed is not just on average associated with the increase in **TARGET**, but this is the case for almost every player. The other coefficients are noisy and not significant, and no clusterization based on the coefficients' values was observed. Also, the average swing speed of the player had no correlation with any of the coefficients of the model, which might imply that the simple model run above generalizes the effects well for each player, but little or no evidence can be provided due to a small number of observations per player. In the future, an additional attempt can be made by carefully adding other pitch types.



4.1.3 Final aggregate model. If all of the variables from Section 4.2.1 are considered, the following model is proposed.

```
glm(formula = TARGET ~ bat_speed_norm + bat_speed_norm_sq + release_speed_norm +
    plate_x_norm_inv + plate_x_norm_inv_less_minus1 + plate_z_norm_norm +
    plate_z_norm_less_0 + diff_hand + pfx_x_inv_norm + pfx_z_norm +
    balls + strikes2 + bat_speed_norm:(release_speed_norm + plate_x_norm_inv +
    plate_x_norm_inv_less_minus1 + plate_z_norm_norm + plate_z_norm_less_0 +
    diff_hand + strikes2), family = binomial, data = d1)
```

	Variable	Coefficient	SE	pvalue
1	(Intercept)	-0.82	0.015	0
2	bat_speed_norm	0.086	0.0038	8.8e-114
3	bat_speed_norm_sq	0.00043	4e-04	0.28
4	release_speed_norm	-0.057	0.003	4.7e-81
5	plate_x_norm_inv	-0.29	0.013	7.1e-118
6	plate_x_norm_inv_less_minus1	0.31	0.025	2.9e-34
7	plate_z_norm_norm	-0.2	0.0051	0
8	plate_z_norm_less_0	0.63	0.094	2.3e-11
9	diff_hand	0.49	0.032	2.4e-54
10	pfx_x_inv_norm	-0.37	0.022	4.5e-64
11	pfx_z_norm	-0.57	0.029	1.9e-84
12	balls	0.04	0.0074	5.4e-08
13	strikes2	0.28	0.017	3.9e-63
14	bat_speed_norm:release_speed_norm	-0.0027	0.00081	0.00088
15	bat_speed_norm:plate_x_norm_inv	0.022	0.0034	2.8e-1

16	<code>bat_speed_norm:plate_x_norm_inv_less_minus1</code>	0.05	0.0065	1.6e-14
17	<code>bat_speed_norm:plate_z_norm_norm</code>	0.0014	0.0014	0.31
18	<code>bat_speed_norm:plate_z_norm_less_0</code>	0.072	0.024	0.0024
19	<code>bat_speed_norm:diff_hand</code>	-0.019	0.0041	2.1e-06
20	<code>bat_speed_norm:strikes2</code>	-0.012	0.0043	0.0042

The coefficients of this model are (some of the variables are a little weird so that I can better interpret the model later):

- `bat_speed_norm`: swing speed normalized by subtracting the batter’s average swing speed (in mph)
- `bat_speed_norm_sq`: square of the normalized swing speed
- `release_speed_norm`: pitch speed normalized by subtracting aggregate average pitch speed in the dataset (in mph)
- `plate_x_norm_inv`: inverted (multiplied by -1 if left stand) normalized horizontal location of the ball
- `plate_x_norm_inv_less_minus1`: interaction term between the inverted normalized horizontal location of the ball and the indicator of the inverted normalized horizontal location is less than -1
- `plate_z_norm_norm`: normalized vertical location of the ball normalized (one more time) by subtracting the average vertical location of the ball
- `plate_z_norm_less_0`: interaction term between the normalized vertical location of the ball and the indicator of the normalized vertical location is less than 0
- `diff_hand`: indicator of the different hand of the pitcher and stand of the batter
- `pfx_x_inv_norm`: inverted horizontal movement of the ball normalized by subtracting the average inverted horizontal movement of the ball
- `pfx_z_norm`: vertical movement of the ball normalized by subtracting the average vertical movement of the ball
- `balls`: number of balls in the at-bat
- `strikes2`: an indicator of the number of strikes in the at-bat is 2
- `bat_speed_norm:release_speed_norm`: interaction term between `bat_speed_norm` and `release_speed_norm`
- `bat_speed_norm:plate_x_norm_inv`: interaction term between normalized swing speed and `plate_x_norm_inv`
- `bat_speed_norm:plate_x_norm_inv_less_minus1`: interaction term between normalized swing speed and `plate_x_norm_inv_less_minus1`
- `bat_speed_norm:plate_z_norm_norm`: interaction term between `bat_speed_norm` speed and `plate_z_norm_norm`
- `bat_speed_norm:plate_z_norm_less_0`: interaction term between `bat_speed_norm` speed and `plate_z_norm_less_0`
- `bat_speed_norm:diff_hand`: interaction term between `bat_speed_norm` speed and `diff_hand`
- `bat_speed_norm:strikes2`: interaction term between `bat_speed_norm` speed and `strikes2`

In the model interpretation, I will only focus on the `bat_speed_norm`. If the ball is pitched at the average speed, in the average location, the pitcher and the batter have the same hand, the movement of the ball is average, the number of balls is 0, and the number of strikes is 2, the effect of the normalized swing speed on the probability of the hit into play is positive, and is very similar to the one in the simple model, so you can refer to the graphs of the fitted probabilities in the simple model to visualize this effect. Moreover, despite the significant interaction terms, the effect of the normalized swing speeds stays positive. For example, let us consider one of the ‘worst-case scenarios.’ If the ball is pitched at 100mph and thrown at the opposite side edge of the strike zone from the pitcher with the opposite hand, the score at bat is 0 balls and 2 strikes, then the coefficient to the `bat_speed_norm` will drop from 0.086 to 0.015, but still will be positive.

Note that you can find my comment about using `xwOBA` as a target (instead of the probability of the hit) in Appendix B.

5 Swing Speed as an Endogenous Variable

In this section, I present the swing speed analysis as an Endogenous Variable. I assume that the environment (the stage of the game) makes the batter adapt his strategy by changing the swing speed (or simply some aspects of the game force him to change the swing speed). I use only aggregate analysis in this section. Since it makes more sense to consider how the swing speed is changed relative to the batter's average, I use the normalized swing speed from the previous section.

Like in the previous section, I present the models and the results in Section 5.1. The results (visualizations and comments) of the exploratory data analysis which helped me to create the variables used in the models can be found in Appendix C.

5.2 Modelling and Results

Before running the final model, I want to mention that the number of outs in the inning and the score in runs are surprisingly insignificant (I tried lots of variables' variations and interactions with the other covariates). Also, the interaction of the final variables in the models did not improve the fit, nor did the use of the lightgbm model instead of the linear regression model.

```
lm(formula = bat_speed_norm ~ release_speed_minus_avg_release_speed_FF +
    avg_release_speed_FF + inning_final + top_inning7plus + balls +
    strikes + b1_loaded + b2_loaded + b3_loaded + plate_x_norm_inv +
    plate_x_norm_inv_less0 + plate_z_norm + plate_z_norm_less1_5 +
    pfx_x:stand:p_throws + pfx_z:stand + sz_bot_demean, data = d1)
```

	Variable	Coefficient	SE	pvalue
1	(Intercept)	9.6	0.5	3.2e-8
2	release_speed_minus_avg_release_speed_FF	-0.19	0.01	8.3e-74
3	avg_release_speed_FF	-0.074	0.0054	7.2e-43
4	inning_final2-5	-0.13	0.034	0.00011
5	inning_final6+	-0.16	0.037	1e-05
6	top_inning7plus	-0.13	0.034	0.00017
7	balls	0.46	0.012	0
8	strikes	-0.89	0.015	0
9	b1_loaded	-0.098	0.025	7.3e-05
10	b2_loaded	-0.17	0.03	2.2e-08
11	b3_loaded	-0.19	0.04	1.2e-06
12	plate_x_norm_inv	-1.6	0.03	0
13	plate_x_norm_inv_less0	2.5	0.054	0
14	plate_z_norm	-0.7	0.0076	0
15	plate_z_norm_less1_5	-0.058	0.037	0.12
16	sz_bot_demean	-2.2	0.34	1.5e-1
17	standL:pfx_z	0.75	0.05	2.8e-51
18	standR:pfx_z	0.9	0.048	8.8e-8
19	pfx_x:standL:p_throwsL	-0.27	0.068	7.7e-05
20	pfx_x:standR:p_throwsL	0.18	0.046	9.1e-05
21	pfx_x:standL:p_throwsR	-0.47	0.051	4.2e-2
22	pfx_x:standR:p_throwsR	-0.1	0.048	0.03

```
[1] "R^2: 0.185"
```

Variables of the model are as follows:

- `release_speed_minus_avg_release_speed_FF`: pitch speed normalized by subtracting the average fastball pitch speed of the pitcher (in mph)
- `avg_release_speed_FF`: average fastball pitch speed of the pitcher (in mph)
- `inning_final`: factor variable of the inning. Three levels: “1” (reference category), “2-5”, “6+”
- `top_inning7plus`: indicator of the top half-inning and the inning is 7 or later
- `balls`: number of balls in the at-bat
- `strikes`: number of strikes in the at-bat
- `b1_loaded`: indicator of the first base loaded
- `b2_loaded`: indicator of the second base loaded
- `b3_loaded`: indicator of the third base loaded
- `plate_x_norm_inv`: inverted (multiplied by -1 if left stand) normalized horizontal location of the ball (from the catcher’s perspective)
- `plate_x_norm_inv_less0`: interaction term between the inverted normalized horizontal location of the ball and the indicator of the inverted normalized horizontal location is less than 0
- `plate_z_norm`: normalized vertical location of the ball
- `plate_z_norm_less1_5`: interaction term between the normalized vertical location of the ball and the indicator of the normalized vertical location is less than 1.5
- `sz_bot_demean`: vertical location of the bottom of the strike zone normalized by subtracting the average vertical location of the bottom of the strike zone of the batter
- `standL:pfx_z`: interaction term between dummy for the left stand and the vertical movement of the ball (from the catcher’s perspective)
- `standR:pfx_z`: interaction term between the dummy for the right stand and the vertical movement of the ball (from the catcher’s perspective)
- `pfx_x:standL:p_throwsL`: horizontal movement of the ball (from the catcher’s perspective) * dummy for the left stand * dummy for the left-handed pitcher
- `pfx_x:standL:p_throwsR`: horizontal movement of the ball (from the catcher’s perspective) * dummy for the left stand * dummy for the right-handed pitcher
- `pfx_x:standR:p_throwsL`: horizontal movement of the ball (from the catcher’s perspective) * dummy for the right stand * dummy for the left-handed pitcher
- `pfx_x:standR:p_throwsR`: horizontal movement of the ball (from the catcher’s perspective) * dummy for the right stand * dummy for the right-handed pitcher

Interpretation: Almost all of the coefficients are of the expected sign. Pitch speed negatively affects the normalized swing speed. The later the innings, the lower the normalized swing speed is on average. Away teams start to swing slower than the home teams after inning 6. The number of balls has a positive association, and the number of strikes has a negative one. Each base loaded decreases the normalized swing speed. The first base has the smallest effect; the third base has the highest one. Keeping everything else constant, the normalized swing speed is highest when the ball is in the middle of the plate (horizontal-wise) and as low as possible vertical-wise. The lower squat of the batter is slightly associated with the decrease in the normalized swing speed. More vertical movement downwards is associated with a lower normalized swing speed. For a right-hand batter, if the pitcher is left-handed, the normalized swing speed is higher if the ball moves away from him horizontally. If the pitcher is right-handed, the effect is the opposite. The left-hand batter swings faster if the ball moves away from him horizontally (this one is unexpected).

6 Conclusion

In conclusion, this report shows the swing speed analysis as both exogenous and endogenous variable. The analysis of the swing speed as an exogenous variable shows that the probability of hit into play is, on average, positively related to the swing speed once we fix the batter. The effect stays positive at almost all feasible values of the other control variables. The coefficient to the squared term is significant in neither of the models presented, suggesting the absence of the optimal swing speed. It can be suggested that the increased swing speed causes an increase in the probability of the hit into play for the batter. However, it can be the case of confounding variables, such as the ‘confidence’ of the batter, which can be affected by the ‘easiness’ of

the pitch. In other words, it is not the case that the batter randomly chooses the swing speed, and increased swing speed causes a better quality of the hit. It is more likely that the batter chooses a higher swing speed because he thinks that he is more prepared for this pitch and feels more confident to swing it at a higher speed. Since he is more prepared, he is more likely to hit the ball into play. Thus, there might be no causal relationship between the swing speed and the probability of the hit into play.

As for the swing speed analysis as an endogenous variable, the batter explains 36% of the variability in swing speed. If we normalize the swing speed, the main predictors of the normalized swing speed are the number of balls, number of strikes, and location of the ball. Note that the model explains only 18.5% of the variability of the normalized swing speed. This means that either there are some other factors missed in the analysis, the noise is relatively high (might be measurement errors), and/or the swing speed is mostly exogenous and is mostly determined by the batter at his discretion, independent of the environment. I think that the first two points are the most likely, and more analysis can be done in future work.

7 Future Work

The analysis reported can be extended further. In fact, this analysis seeds more questions than answers. A more detailed causal inference analysis can be done for the exogenous part of the analysis. At the same time, a more robust analysis can be created to model the quality of the hit for each batter individually. Furthermore, other measures of the quality of the hit can be considered (i.e., xWOBABatter). I have also done probability of contact (any contact with the ball). The results were pretty much the same as for the probability of getting hit into play. Analyzing all pitch types (not just fastballs) might help with robustness in both parts of the analysis. As for the endogenous part of the analysis, more variables can be derived to improve the model (i.e., the autocorrelation of the swing speed per batter, a measure of the tiredness of the batter, and an analysis of the extreme values of the normalized swing speed). At the same time, more research can be done to look for any current analysis on the swing speed (since it is new data, there might be a new analysis/information every day). More qualitative analysis can be done. For example, I would like to find and ask people related to baseball what makes them swing at different speeds. Also, I could include broader aspects of the bat tracking data (i.e., bat length, which is also newly available data). Finally, it could be useful to consider not only proper swing data and merge the current analysis with the swing decision model since the factors affecting the decision to swing (which I might miss in this analysis) might similarly affect the swing speed.

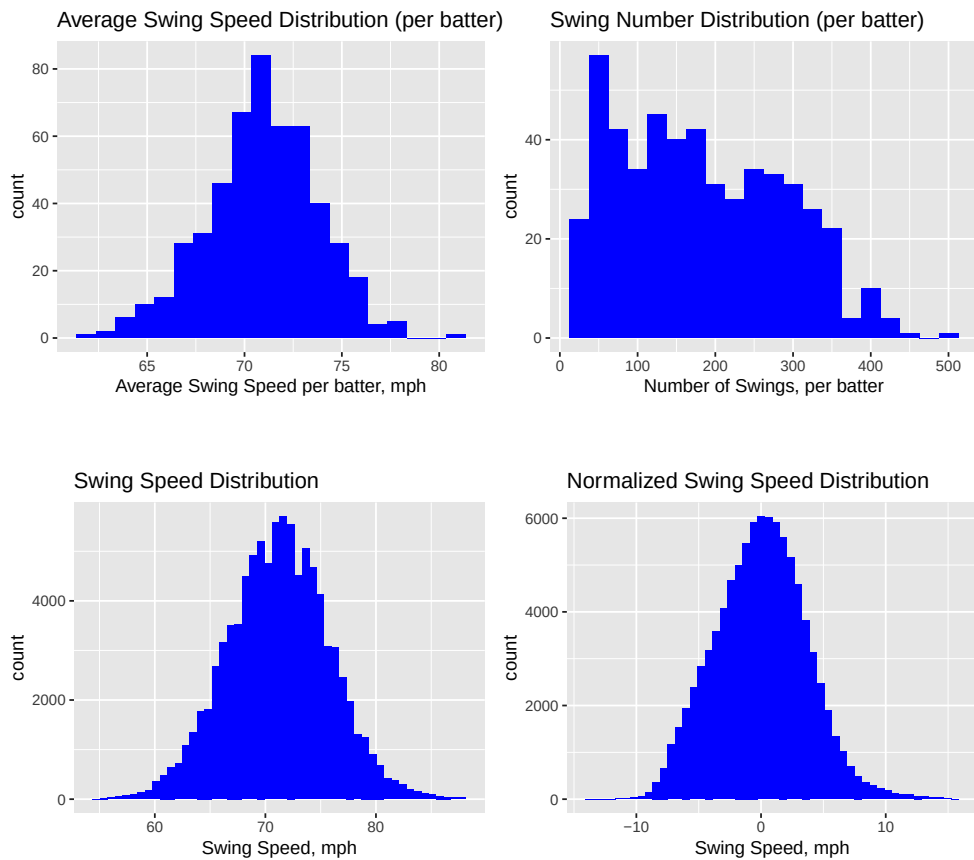
Appendix

Appendix A: Exploratory Data Analysis, exogenous analysis.

To begin with, my target variable for the exogenous analysis is the binary variable, which equals one if the batter hits the ball into the play and zero otherwise. The variable is called **TARGET**. In the cleaned data, 34% of the swings resulted in the hit into play.

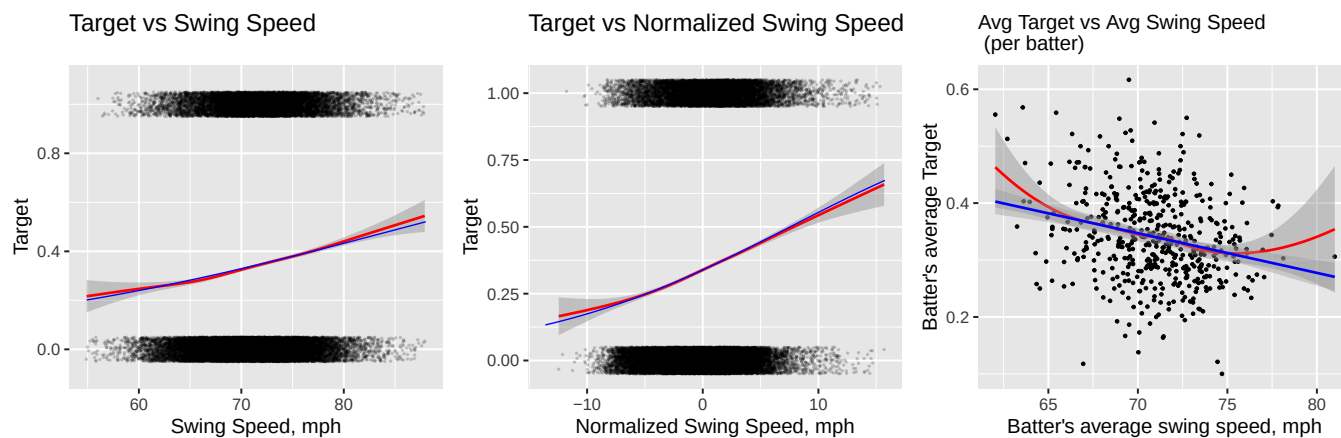
In this subsection, I will present the variables which are correlated with the **TARGET** variable, which will be used in the modelling.

Let me start with the swing speed. In the first histogram below, you can see the variability in the average swing speed of the batters. To be able to conduct a more interpretable aggregate analysis (with all of the players at the same time), I created the normalized swing speed variable.

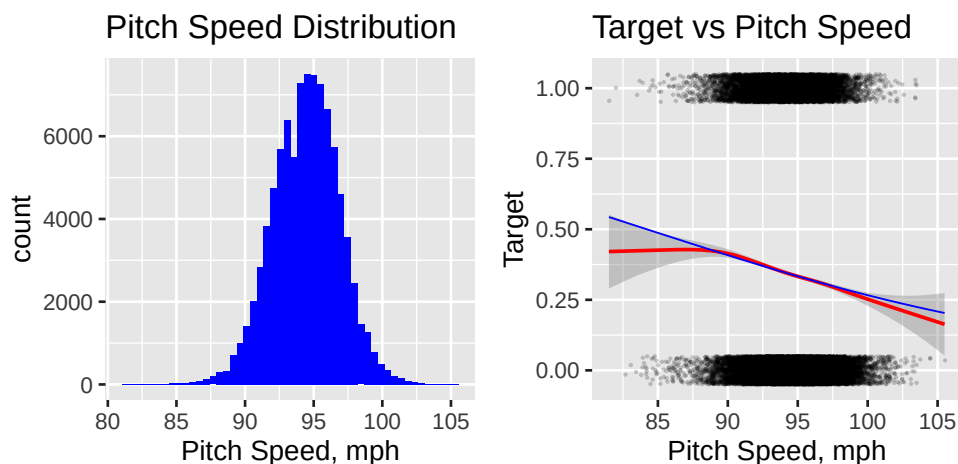


The player with the higher average swing speed tends to have less proportion of balls hit into play. However, the relationship between **TARGET** and normalized swing speed is positive. This means that if we fix some batter, then on average, the higher swing speed is associated with the higher probability of hitting the ball into play. Moreover, the latter effect dominates so that, in general, the swing speed is associated with the higher probability of hitting the ball into play even if we do not fix some batter.

At the same time, no apparent concavity is noticed at this point, so it might not be the case that the optimal swing speed exists.

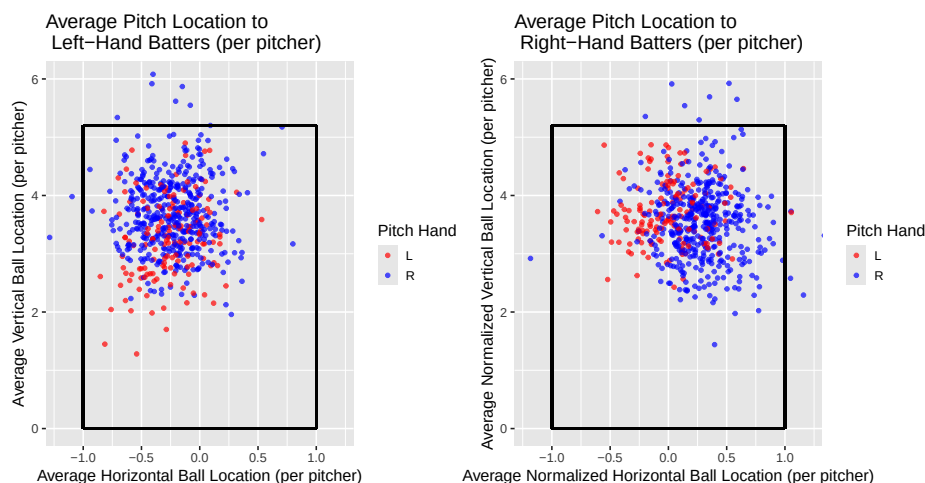


The fastball's release speed (the pitch speed) is more stable between the pitchers than the swing speed between the batters. As expected, the release speed is negatively associated with the probability of hitting the ball into play.



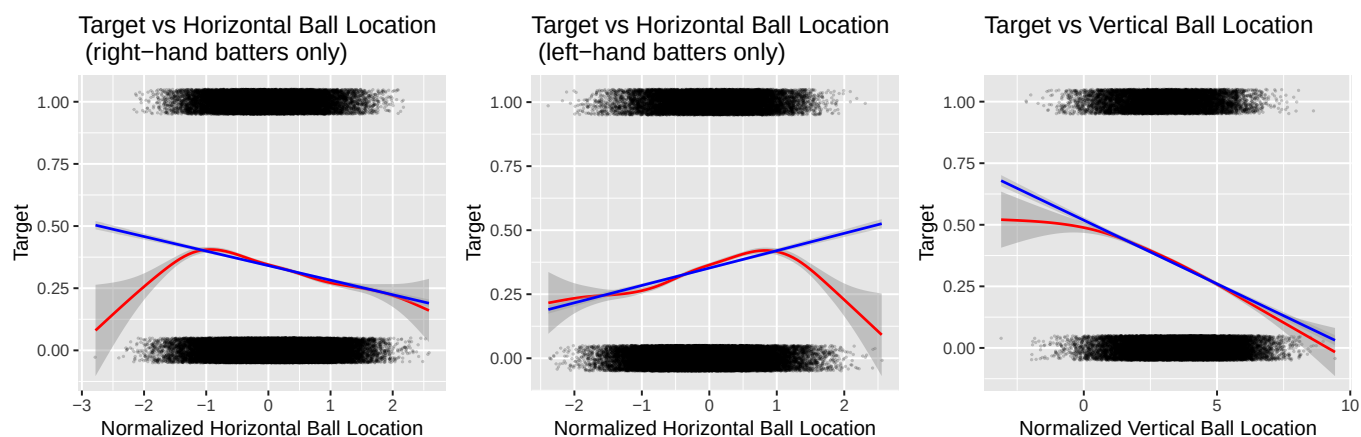
The location of the ball (the point where it crosses the strike zone) heavily depends on the pitcher's throwing hand and the batter's stand. The diagrams below represent the strike zone from the catcher's perspective. Thus, it can be seen that, on average, both left-hand and right-hand batters receive the ball on the opposite side of the strike zone. At the same time, right-hand and left-hand pitchers throw the balls to the different regions of the strike zone when pitching to right-hand batters. Thus, it is important to look at each of these four cases separately (l to r, r to r, l to l, r to l) before determining the association between the location of the ball and the probability of hitting the ball into play.

Note that in the diagrams below and further, I refer to the ball's 'Normalized Vertical/Horizontal' location. The normalized version of the location of the ball is derived such that the normalized horizontal location of the left edge of the strike zone (from the catcher's perspective) is equal to -1, and the right edge is equal to 1, and the normalized vertical location of the bottom edge of the strike zone is equal to 0, and the top edge is equal to 5.2 (this weird 5.2 value is because the height of the strike zone is, on average, 2.6 higher than the width of the strike zone).

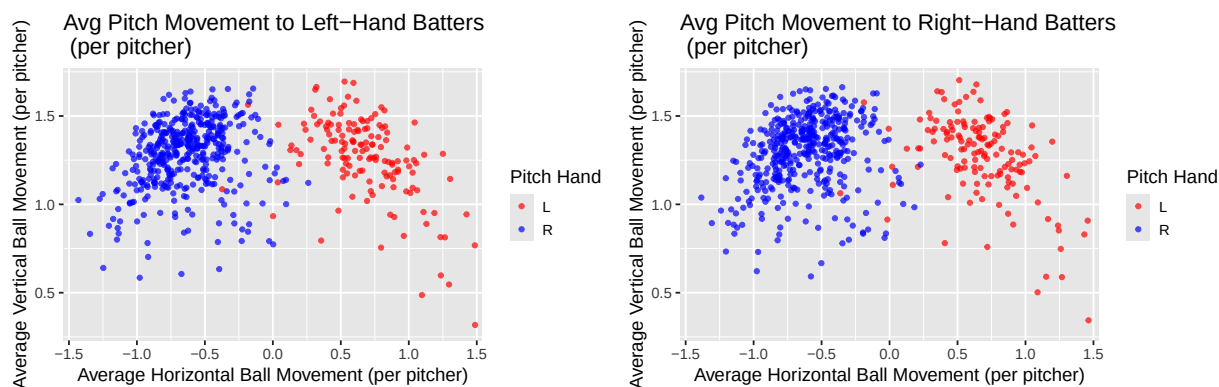


After looking at each of the 4 cases, I conclude that the hand of the pitcher does not affect the effect of the ball's location. However, the stand affects how the horizontal location of the ball affects the probability of

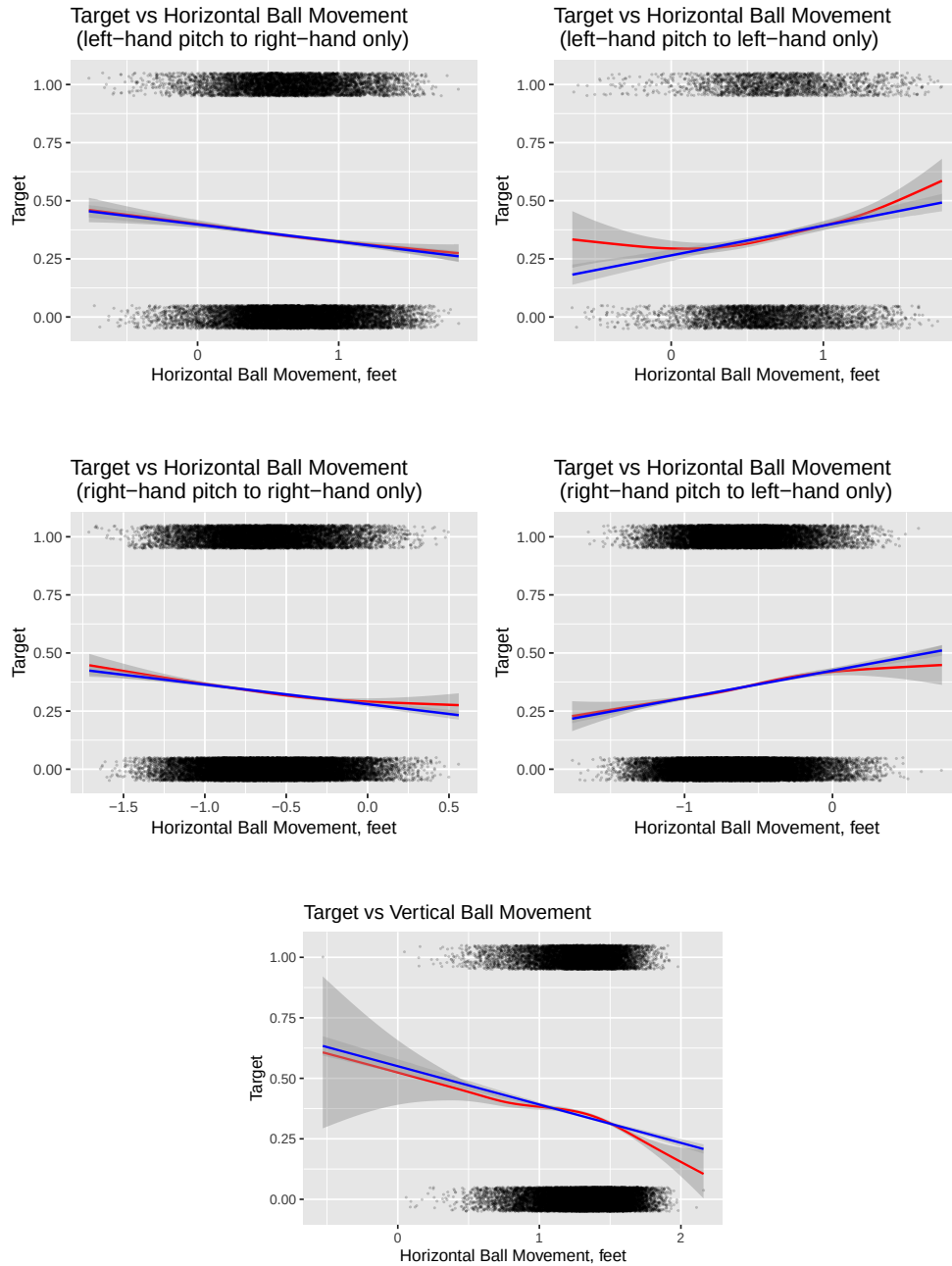
hit into play. We can see that if the ball is thrown in the strike zone, the further it is from the batter, the less likely the batter hits the ball into play. The effect is the opposite once the ball is thrown outside the strike zone to the batter's side. If we inverted the x-axis for the left-hand batters, we would get almost identical plots for the left-hand and right-hand batters. Thus, I propose to invert (multiply by -1) the horizontal location of the ball for the left-hand batters and use this inverted variable in the modeling. As for the vertical location of the ball, the effect is the same in each case. As long as the ball is not thrown below the strike zone, the higher the ball, the less likely the batter hits the ball into play.



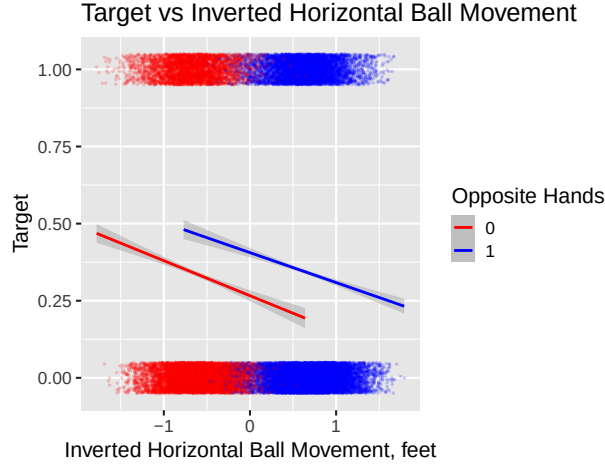
The vertical movement of the ball is, on average, the same for the left-hand and right-hand batters. The horizontal movement is inverted ('mirrored') for the left-hand batters, which is expected.



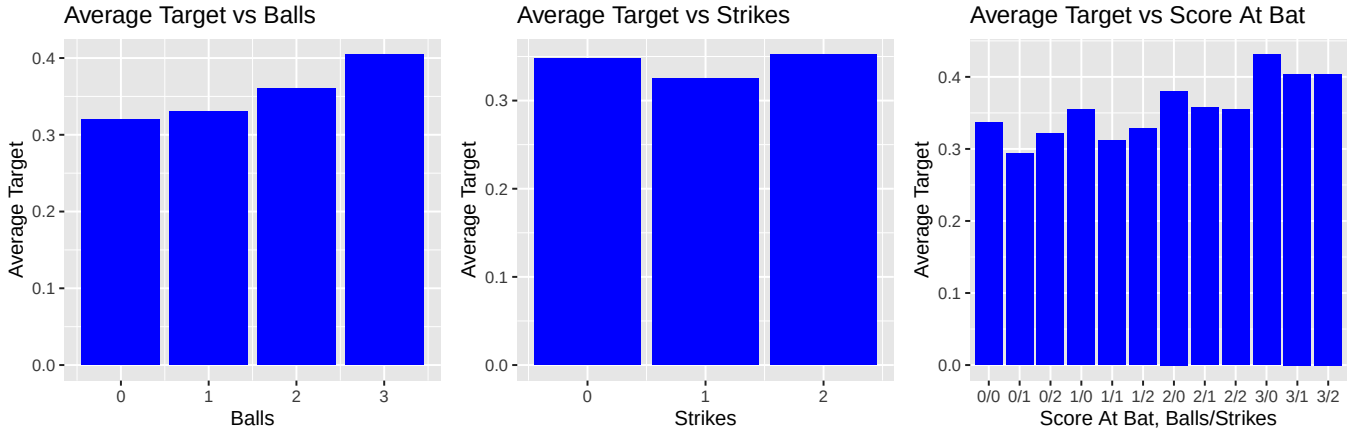
For the horizontal movement, the hand of the pitcher and the stand of the batter are the same, then the further the ball moves from the batter, the less likely he hits the ball into play. If the hand and stand are opposite, the effect is the opposite. Regardless of the hand and stand, the vertical movement has a stable negative association with the TARGET.



Like for the horizontal location of the ball, I created the inverted variable for horizontal movement (multiply by -1 if the stand is left). The relationship between this inverted variable and hit into play for the same hand and stand and opposite hand and stand cases turns out to be similar. The slopes are the same, so I only need to add the dummy variable indicating whether the stand and pitch are the same (or different).



The number of balls in at bat is positively associated with the **TARGET**, while the association is negative for the number of strikes. As the number of balls increases, the pitchers become less risky, and at the same time, the batter does not rush and waits for an ‘easy ball,’ which results in a higher probability of getting hit into play. Linearity is observed for balls, while the unexpected jump at two strikes is observed for strikes. This might be because the batter becomes very risk-averse and tries to hit the ball with much less power. Otherwise, it is hard to explain such phenomena.



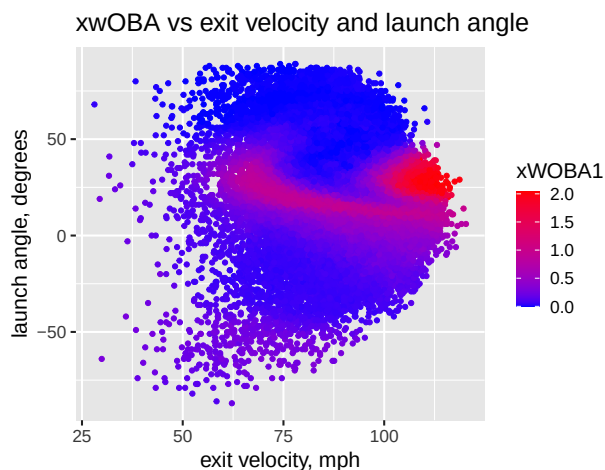
Appendix B: A couple of words about xWOBA as a measure of the quality of the hit.

The main drawback of using the probability of hit into play as a measure of the batter’s performance is that it treats any hit as a success. For example, home run and easy-to-catch pop-ups are treated similarly. We can use WOBA (weighted on-base average) weights to distinguish between these cases. wOBA is the measure introduced by Tom Tango. In its [formula](#), each outcome of the pitch (i.e. first base, home run) is multiplied by some coefficient that can be interpreted as the quality of the batter’s performance at a certain pitch. So for each pitch the corresponding wOBA weight can be attached depending on the outcome.

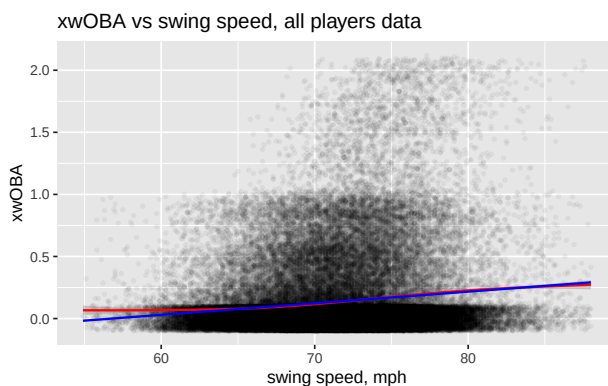
Such corresponding wOBA weight will depend not only on the batter’s performance (i.e., how good the ball was hit), but also on the performance of the fielders (i.e., the fielders can make an error and allow the batter to advance to the further base, which would result in a larger wOBA weight attached). To eliminate the effect of the fields, Statcast introduced [Expected wOBA](#), aka **xwOBA** which mainly depends on the exit

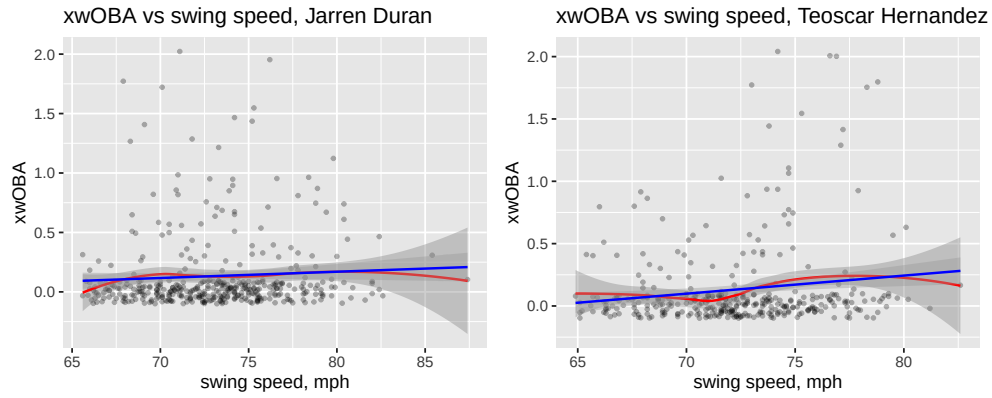
velocity and launch angle. As a result, xwOBA can be interpreted as the “pure measure of the batter’s performance at the pitch.”

I initially used xwOBA as a target variable in the analysis above. However, xwOBA turned out to have quite an unstable distribution. Most of the values in the data are zero, with some rare large values. This can be explained by the fact that xwOBA has significant positive values only at selected exit velocity and launch angle values. Also, xwOBA is not quite a monotonic or smooth function of exit velocity and launch angle. As a result, the analysis above using xwOBA as the target variable resulted in unstable results, so I chose a less suitable but more stable probability of the hit instead.



The only consistent (but not necessarily significant) findings are that xwOBA on average increases with the swing speed (in general and for most players individually). In addition, xwOBA is almost always upper-bounded by 1 until the bat speed reaches a threshold in the range of 66 and 72 mph. xwOBA starts taking values more than 1 once this threshold is reached. The upper-bound becomes 2 (maximal possible value when it’s very likely that the hit results in a home run) very quickly. It applies both to the aggregate and individual data.

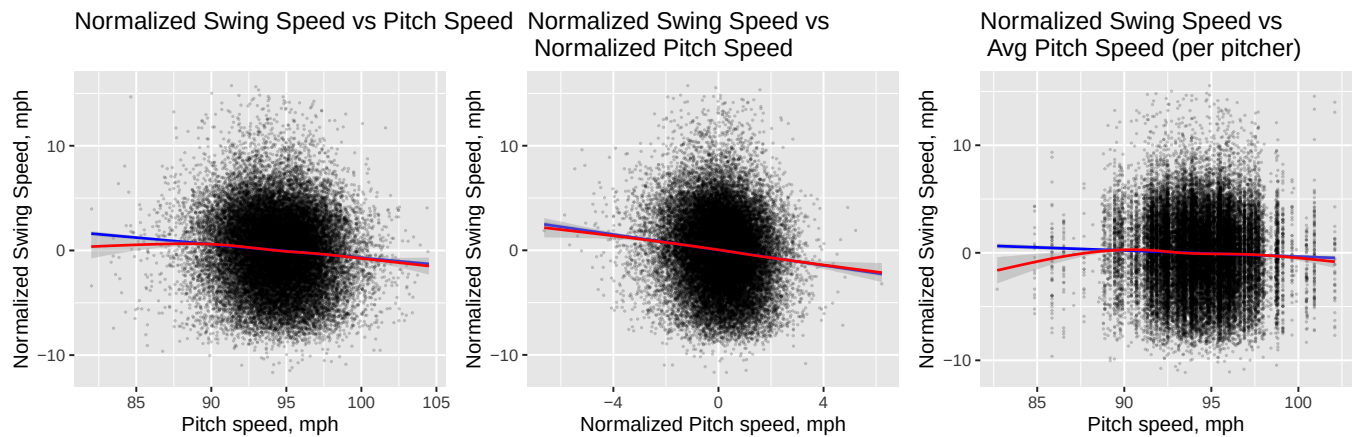




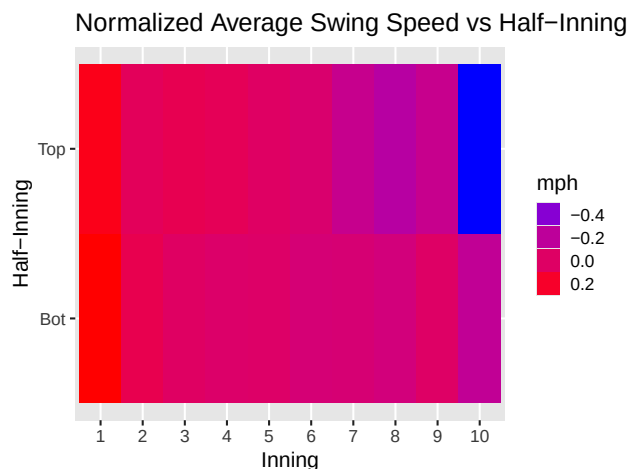
In further analysis, more data and more robust techniques can be used to consider xwOBA as the target for the analysis above.

Appendix C: Exploratory Data Analysis, endogenous analysis.

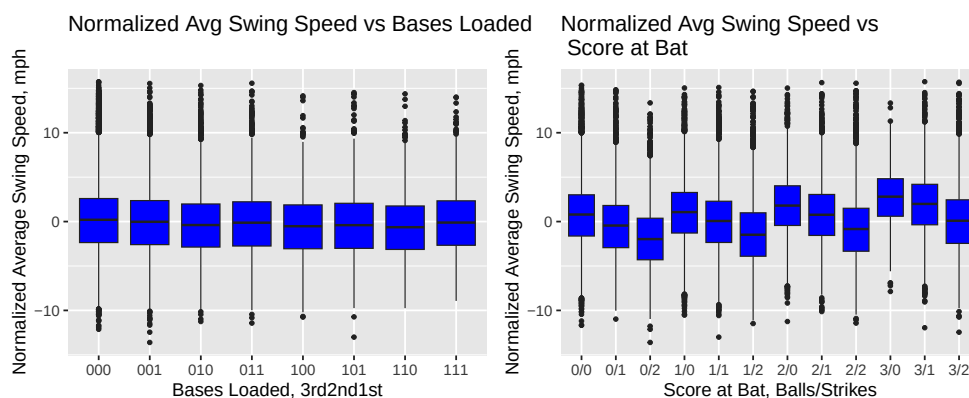
There is a negative relationship between normalized swing speed and pitch speed. This effect can be split into two parts. There is a slight negative effect on the pitcher's average pitch speed of the pitcher, and the negative effect on the pitch speed relative to the pitcher's average.



Later innings tend to decrease the normalized swing speed gradually. The away team (bats during the top half-inning) starts to swing even slower than the home team in the later innings.

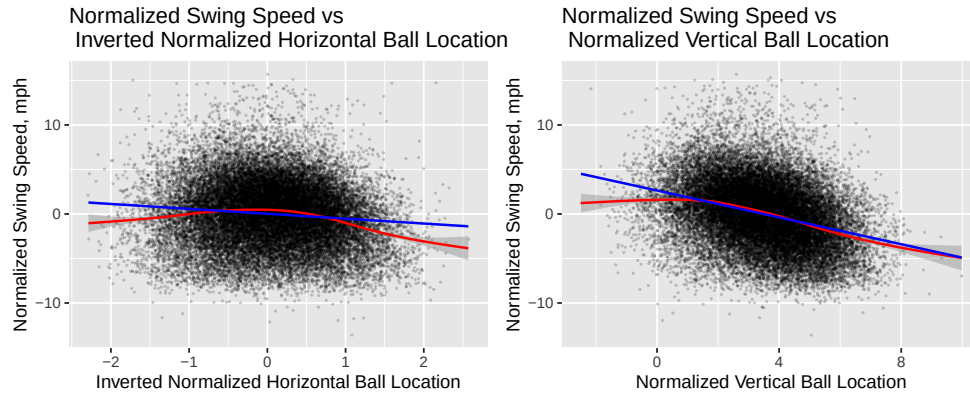


Each base loaded has a relatively constant effect on the normalized swing speed, no matter how the other bases are loaded. Score at bat is one of the most significant variables. The number of balls is positively related to normalized swing speed, while the number of strikes is negatively related to normalized swing speed.



The ball's location is also one of the best predictors of the normalized swing speed. If the normalized horizontal location of the ball is inverted, then the relationship becomes the same in each of the 4 cases (right-hand pitcher throws to right-hand batter, l to l, r to l, and r to r), so there is no need to separate. The slope of the smooth regression line changes at 0. Similarly, the slope is changed at the value of 1.5 for the normalized vertical location.

Remember that the normalized version of the location of the ball is derived such that the normalized horizontal location of the left edge of the strike zone (from the catcher's perspective) is equal to -1, and the right edge is equal to 1, and the the normalized vertical location of the bottom edge of the strike zone is equal to 0, and the top edge is equal to 5.2 (this weird 5.2 value is because the height of the strike zone is, on average, 2.6 higher than the width of the strike zone).



In all four cases, horizontal and vertical movement did not seem significant in bivariate analysis. However, these variables showed significance in the modeling.

Similarly, the location of the bottom of the strike zone (normalized relative to the average location of the bottom of the batter's strike zone) is significant. I thought this variable would be significant because I thought that how the batter bends his legs might affect the normalized swing speed, which can be proxied by the vertical coordinates of the top and bottom of the strike zone. The location of the top of the strike zone (normalized) is not significant, though.